

Probability Density Function and the CDF of a Continuous Variable

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).



- **Builds on:** Probability distributions ([Note MA-003](#)); Probability mass function (PMF) ([Note MA-021](#)); Expected value and variance of a random variable ([Note MA-012](#)).
- **Leads to:** Density estimation ([Note MA-023](#)); Normal distribution ([Note MA-024](#)); Standard normal and the z-table ([Note MA-025](#)); P-values ([Note MA-041](#)); Exponential distribution ([Note MA-071](#)).
- **Compare with:** Frequency tables ([Note MA-007](#)); Bernoulli and binomial distributions ([Note MA-021](#)); Pareto distribution and power laws ([Note MA-030](#)); Likelihood ([Note MA-069](#)).

1. Overview

Key point: For a continuous random variable the probability of any single exact value is 0, so the height of the PDF (probability density function) is a density, and probabilities are areas under it; the CDF (cumulative distribution function) is the area to the left of x .

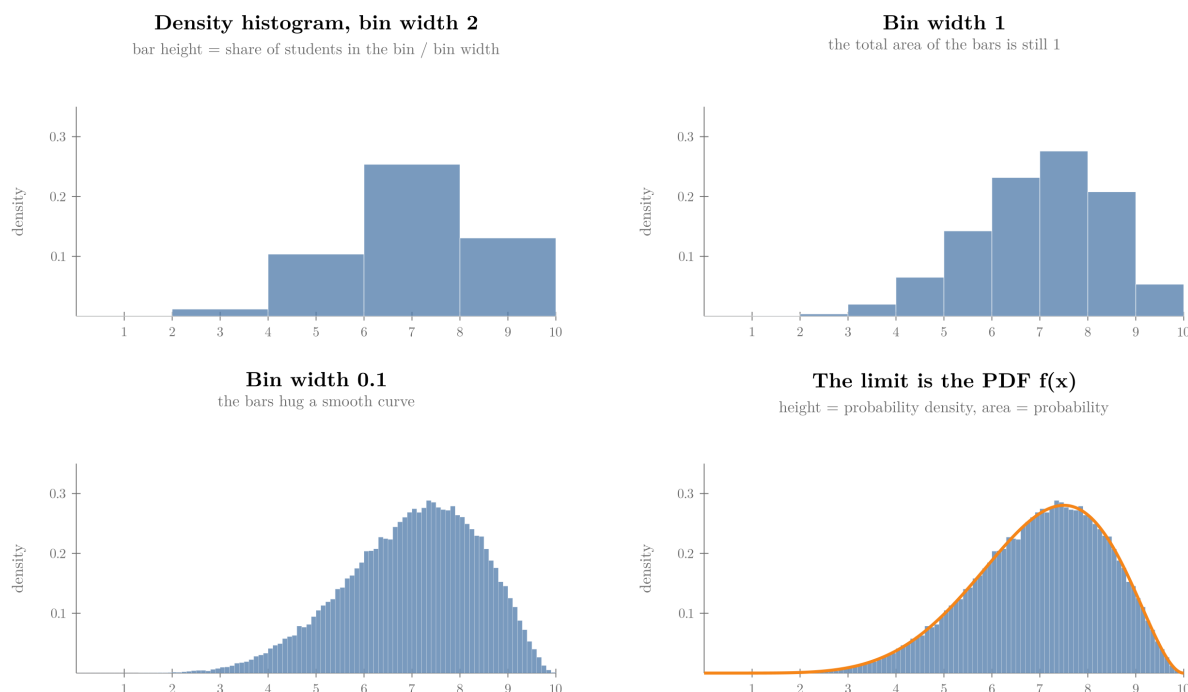


Figure 1: Density histograms with narrower and narrower bins settle onto the PDF curve

Figure 1 shows where the PDF comes from. We take 100,000 CGPAs and draw a density

histogram, in which each bar's **area** is the share of students in its bin. As the bins get narrower, the bars hug a smooth curve: the probability density function.

We met this curve in [the density plot](#) as the KDE (kernel density estimate, a smooth curve fitted to the data) drawn over a histogram, where we noted that its height is a density, not a probability. This Note goes step by step:

- why the probability of one exact value is 0 (section 2);
- why probabilities are areas under the curve (section 3);
- why the height is a density, and what a PDF is (sections 4 and 5);
- the CDF of a continuous variable, and its link to the PDF (sections 7 and 8).

How the curve is estimated from data is the topic of [density estimation](#).

2. The probability of one exact value is 0

Key point: A continuous variable has infinitely many possible values, so the probability of any one exact value is 0; probabilities only exist for ranges.

The curve in Figure 1 describes the CGPAs of students, a **continuous random variable** (G-466): it can take any value from 0 to 10, with every decimal in between (see [discrete and continuous random variables](#)). The curve peaks at a CGPA of 7.5, where its height is 0.280. Is 0.280 the probability that a student's CGPA is exactly 7.5? It is tempting to read it that way, as we read a bar chart of a PMF (probability mass function: the bar chart of a discrete variable, whose bar heights are probabilities; see [the probability mass function](#)). The answer is no.

Pick one student and ask the probability that their CGPA is exactly 7.912. Not 7.9121, not 7.91199: exactly 7.912, to the last of infinitely many decimals. Between 0 and 10 there are infinitely many possible values. Spreading a total probability of 1 over infinitely many values leaves 0 for each one:

$$P(X = 7.912) = 0$$

The same shows up in real data. Among 100 students, perhaps one has a CGPA of 7.912 to three decimals. Ask for 7.912959 and almost certainly nobody has it; the finer we ask, the closer the share gets to 0.

Shrinking a range shows it with numbers. For the CGPA curve, the probability of a CGPA between 8 and $8 + h$ falls to 0 as the width h shrinks:

Width h	$P(8 \leq X \leq 8 + h)$	$P(8 \leq X \leq 8 + h) / h$
1	0.2088	0.2088
0.1	0.0261	0.2607
0.01	0.00264	0.2639
0.001	0.000264	0.2642

So a graph of “probability at each x ” would be flat at 0 everywhere and tell us nothing. The last column, however, settles on a number: 0.264. The settled value, 0.264, is the density at 8 (section 5). Figure 2 animates the table: as the strip at 8 narrows, watch the orange probability sink to 0 while the blue ratio flattens onto the dashed line at 0.264.

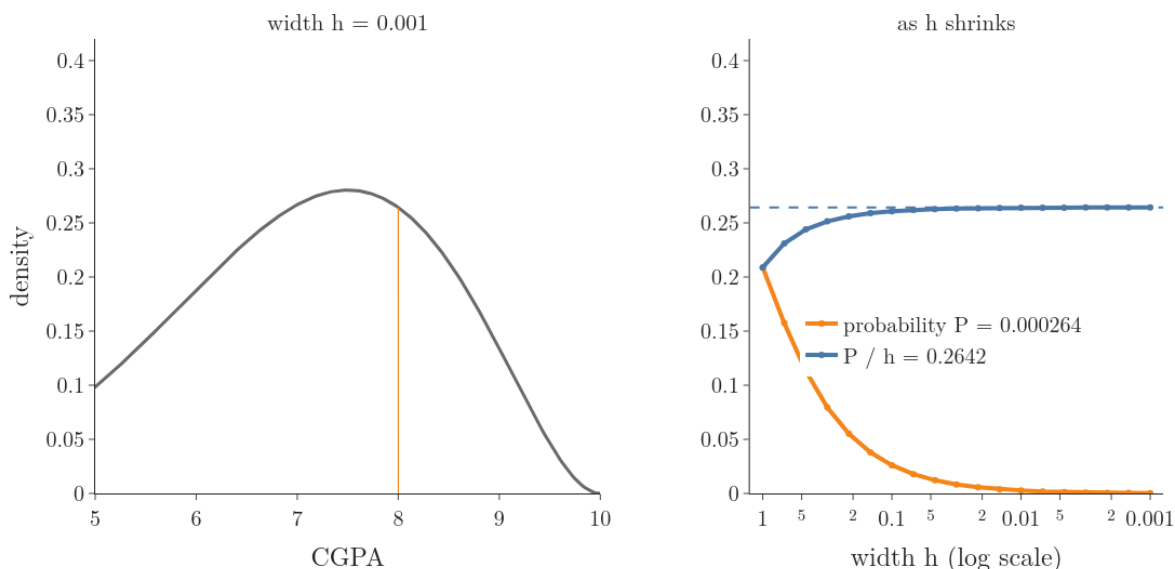


Figure 2: A strip from 8 to $8 + h$ narrows from $h = 1$ to $h = 0.001$: its probability goes to 0, probability divided by width settles at $f(8) = 0.264$

3. Area under the curve is probability

Key point: The total area under a PDF is 1; the area between two values a and b is the probability that the variable falls between them.

A single value has probability 0, so we ask about a range instead. “A CGPA of about 8” becomes “a CGPA between 7.9 and 8.1”. A range has a width, so the curve above it encloses an area, and that area is the probability: 0.0528 for 7.9 to 8.1 (computed in the Python box of this section). A single value is a line with no width, so its area, and its probability, is 0.

The whole area under the CGPA curve stands for the probability that a student’s CGPA is somewhere between 0 and 10. A CGPA somewhere in that range is certain, so the **total area under every PDF is 1**, just as the bars of a PMF add up to 1.

A wider slice gives a larger probability. The area between 8 and 9 is the probability of a CGPA between 8 and 9 (Figure 3, left). Since the curve is not a rectangle, the area is found by **integration** (G-957), which adds up the area of infinitely many infinitely thin strips under the curve.

1. **In words:** the probability that X falls between a and b is the area under the PDF from a to b .

2. **Example:** for the CGPA curve, the area from 8 to 9 is 0.209 (the Python box below finds it by adding up thin strips: `integrate.quad`). About 21% of students have a CGPA between 8 and 9.

3. **Formula:**

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

The symbol $\int_a^b$ reads “the area from a to b under”, and dx marks x as the variable along the horizontal axis. For the CGPA curve:

$$\begin{aligned} P(8 \leq X \leq 9) &= \int_8^9 f(x) dx \\ &= 0.209 \end{aligned}$$

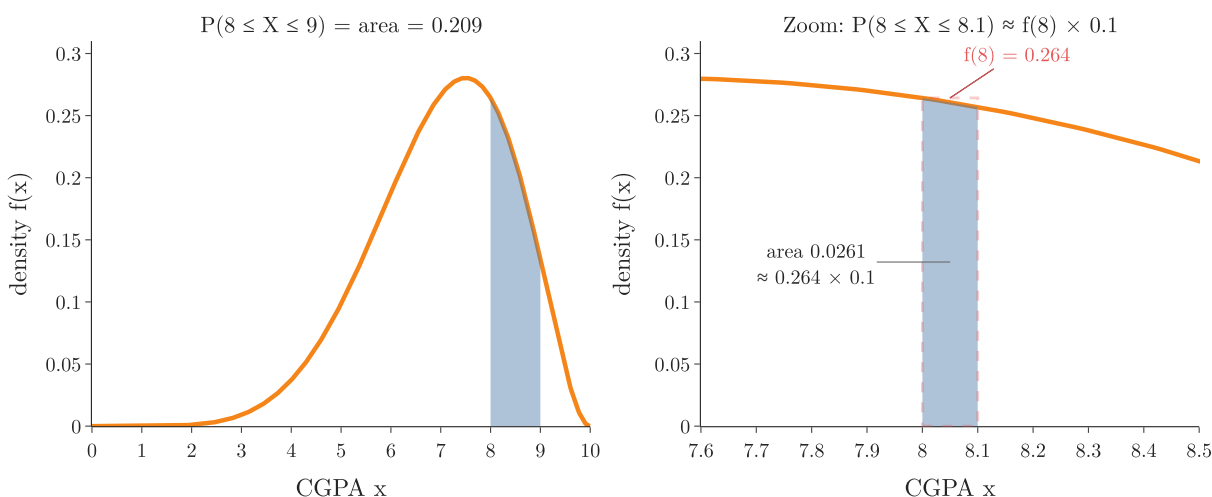


Figure 3: Probability is area under the PDF: between 8 and 9 (left), and a thin slice from 8 to 8.1 (right)

Because a single point has zero width, it has zero area. So for a continuous variable it makes no difference whether the ends are included:

$$P(8 \leq X \leq 9) = P(8 < X < 9)$$

Python: The area from a known PDF, two ways.

```
from scipy import stats, integrate
```

```
# CGPA curve used in this Note: 10 x Beta(7, 3)
```

```
cgpa = stats.beta(7, 3, scale=10)
```

```
# area by numerical integration
```

```
integrate.quad(cgpa.pdf, 8, 9)[0] # 0.2088
```

```
# the same area from the CDF (section 7)
```

```

cgpa.cdf(9) - cgpa.cdf(8)           # 0.2088
cgpa.cdf(8.1) - cgpa.cdf(7.9)      # 0.0528
cgpa.pdf(7.5)                       # 0.280, the peak height

```

The curve is a scaled beta distribution, chosen because it lives exactly on 0 to 10; any smooth curve with area 1 would do.

4. Bars whose area is probability: the PDF

Key point: If each bar's area, not its height, is the probability of its bin, the heights stay put as the bins narrow and trace a curve; that height is a probability density, and the curve is the PDF.

Where does the curve come from, and what is its height? Start from data: the 100,000 CGPAs, cut into bins. There are two ways to draw a bar for each bin, and Figure 4 draws both side by side as the bins get narrower.

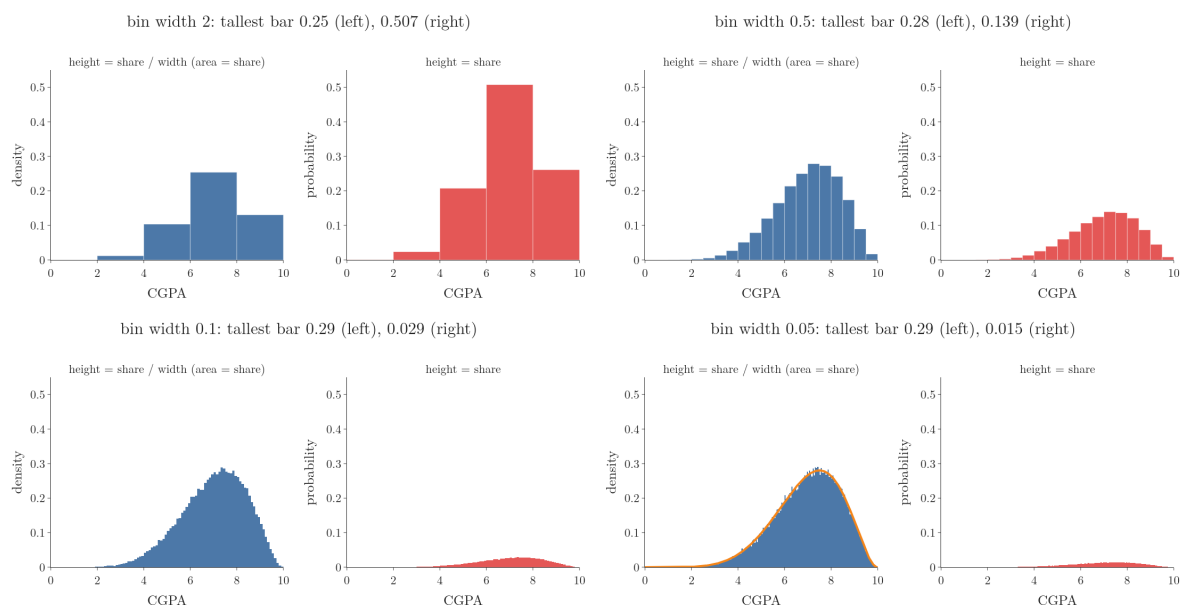


Figure 4: The same CGPAs in narrower and narrower bins. Left: bar area = share of students, the heights stay put and settle onto a curve. Right: bar height = share, every bar sinks towards 0 (idea after 3Blue1Brown, “Why ‘probability of 0’ does not mean ‘impossible’”)

1. **Height = probability (right panel).** Each bar's height is the share of students in its bin. Halve the bin width and each bin holds about half as many students, so every bar roughly halves: the tallest bar falls from 0.507 at width 2 to 0.015 at width 0.05. In the limit, all bars sink to a flat line at 0, the “probability at each x ” of section 2, which tells us nothing about the shape.
2. **Area = probability (left panel).** Each bar's height is the share divided by the bin width, so its **area**, height times width, is the share. Halve the width and the share roughly halves too, so the height stays about the same: the tallest bar stays between

0.25 and 0.29 at every width. The bars keep the shape and settle onto a smooth curve. At every width, the total area of the bars is 1.

The height on the left is “probability divided by width”: a probability **per unit** of CGPA. It is called a **probability density** (G-1569). The curve the bars settle onto is the **probability density function (PDF)** (G-1568): the probability distribution function of a continuous random variable. Its graph is an unbroken curve, and the probability of any range is the area under it, as in section 3.

The x axis works as for a PMF: it holds the values of the variable. The difference is the y axis:

	PMF	PDF
Random variable	discrete	continuous
y axis	probability	probability density
Probability of a range	sum of the bar heights	area under the curve

The second row is the one to remember. On a PMF we read probabilities straight off the graph. On a PDF the height is a density. Figure 5 puts the two side by side: bars we can read as probabilities, and a curve whose height we cannot. The [density plot](#) already met this: its curve’s height is a density, and probability is the area under it.

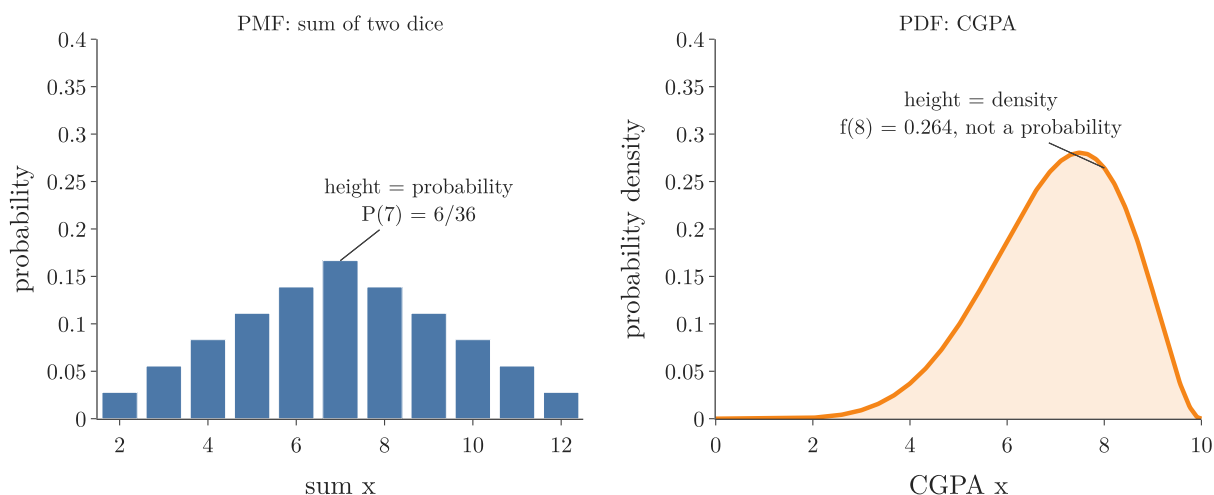


Figure 5: A PMF (sum of two dice, left) and a PDF (CGPA, right): the bar heights are probabilities, the curve’s height is a density

5. What the density at a point means

Key point: The density at x is the probability per unit of x near x : over a narrow width h , the probability is about $f(x) \times h$.

The density $f(x)$ is what probability turns into when we divide by the width of a narrow range (the last column of the table in section 2). Turned around, a thin strip under the curve is almost a rectangle of height $f(x)$ and width h (Figure 3, right).

1. **In words:** the probability of landing in a narrow range is about the density times the width of the range.
2. **Formula:** for a small width h :

$$P(x \leq X \leq x + h) \approx f(x) \times h$$

3. **Example:** the density at 8 is $f(8) = 0.264$. For the range 8 to 8.01:

$$\begin{aligned} P(8 \leq X \leq 8.01) &\approx 0.264 \times 0.01 \\ &= 0.00264 \end{aligned}$$

The exact area is 0.002639; the smaller h , the better the match.

So for practical purposes a higher curve still means “more likely around here”, and we can compare densities at two points the way we compare probabilities. Strictly, though, the height is never a probability itself.

The density histogram of Figure 1 rests on the same idea. A density bar has height “share of the data in the bin divided by the bin width”, so its area is the share. As the bins narrow, those heights become $f(x)$.

Extra: Because a density is probability **per unit**, it can be larger than 1 (see [the normal densities in Gaussian Naive Bayes](#)). A uniform distribution on 0 to 0.5 has height 2 everywhere: width 0.5 times height 2 gives the required area of 1. A probability can never exceed 1; a density can.

6. Famous PDFs

Key point: The normal and log-normal distributions are famous continuous distributions with known PDF formulas; Poisson, often listed with them, is discrete.

The famous continuous distributions of [the list of famous distributions](#) (Figure 6) each have a PDF formula:

- **Normal distribution** (G-1343): parameters μ (mean, location) and σ (standard deviation, scale). Much natural data follows it. Its PDF, worked through in [Gaussian Naive Bayes](#), is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where $\pi \approx 3.14$ and $e \approx 2.718$. For $\mu = 0$ and $\sigma = 1$ at $x = 0$, the exponent is 0 and $e^0 = 1$:

$$\begin{aligned} f(0) &= \frac{1}{\sqrt{2\pi}} \\ &= 0.399 \end{aligned}$$

This is the peak of the normal curve in Figure 6, left.

- **Log-normal distribution** (G-1115): takes only positive values, with a hump near the left and a long right tail (Figure 6, middle). A variable X is log-normal when its logarithm is normal: take a normal Z and set $X = e^Z$. For example, $Z = 0$ gives $X = 1$ and $Z = -1$ gives $X = 0.37$; any Z gives a positive X . Its parameters are also μ and σ , those of the normal $Z = \log X$ (SciPy `lognorm` docs).

The **Poisson distribution** (G-1509), with parameter λ , is often listed beside them, but it counts events $(0, 1, 2, \dots)$, so it is discrete and has a PMF, not a PDF. SciPy, for example, gives `poisson` a `pmf` and no `pdf`. Each of these distributions gets its own Note later. Figure 6 shows one member of each family, with parameters picked only for illustration: two smooth curves, and one set of bars.

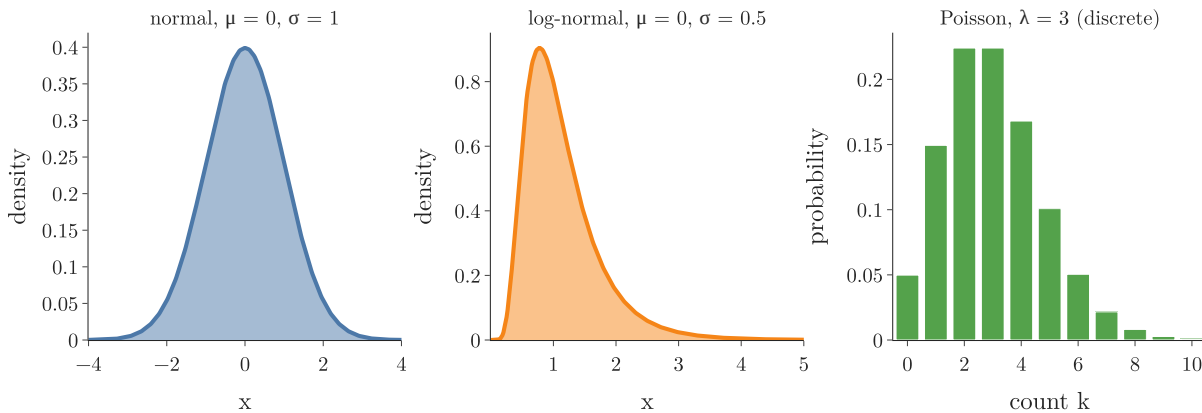


Figure 6: Normal (mean 0, standard deviation 1) and log-normal ($\mu = 0, \sigma = 0.5$) PDFs, and the Poisson PMF with $\lambda = 3$

7. The CDF of a continuous variable

Key point: The CDF $F(x) = P(X \leq x)$ is the area under the PDF to the left of x ; it rises smoothly from 0 to 1.

The **cumulative distribution function** (CDF, G-515) has the same definition as for a discrete variable (see [the CDF of a discrete variable](#)): the probability of a value at most x . For a continuous variable, “at most x ” is the area under the PDF from the far left up to x .

1. **In words:** the CDF at x is the whole area under the PDF to the left of x .
2. **Formula:**

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

The letter t runs along the axis up to x ; $-\infty$ means “from the far left”.

3. **Example:** adult heights in a group follow a normal distribution with mean 165 cm and standard deviation 10 cm. Half the area lies left of the mean, so $F(165) = 0.5$: half the people are 165 cm or shorter. The area left of 150 is $F(150) = 0.067$: about 7% are 150 cm or shorter.

Figure 7 puts the two functions one above the other. The same point means different things on each:

- **On the PDF at 165:** the height $f(165) = 0.0399$ is a density. Roughly, heights near 165 cm are the most common ones.
- **On the CDF at 165:** the height $F(165) = 0.5$ is a probability: the chance of 165 cm or less.

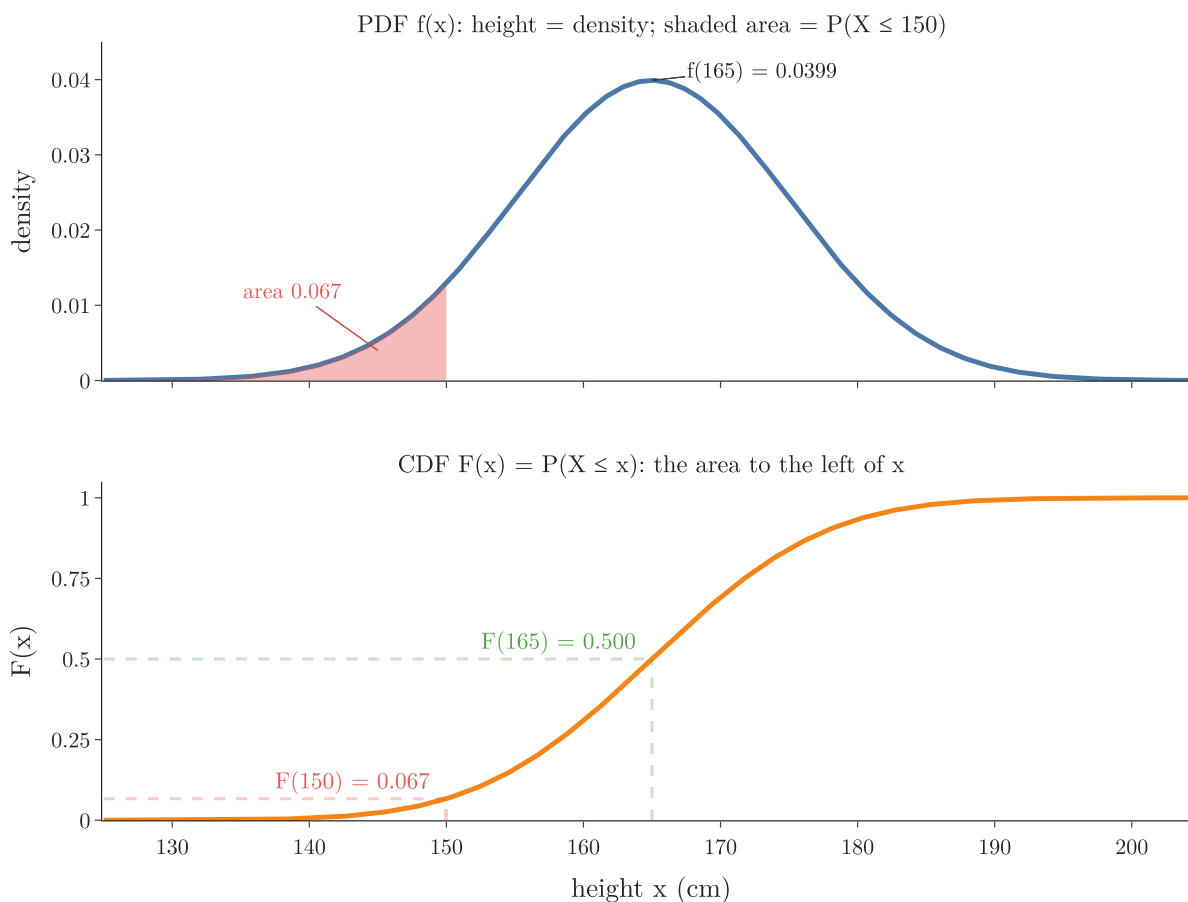


Figure 7: Heights: the PDF with the area left of 150 shaded (top) and the CDF that reads off such areas (bottom)

The CDF of a continuous variable has no steps: it rises smoothly, steepest where the PDF is highest.

Python: PDF and CDF values.

```
height = stats.norm(165, 10)
height.pdf(165)    # 0.0399, a density
height.cdf(165)    # 0.5, a probability
height.cdf(150)    # 0.067
```

Extra: The CDF turns any range into a subtraction:

$$P(a < X \leq b) = F(b) - F(a)$$

Heights between 155 and 175 cm, one standard deviation either side of the mean:

$$P(155 < X \leq 175) = F(175) - F(155)$$

$$P(155 < X \leq 175) = 0.841 - 0.159$$

$$P(155 < X \leq 175) = 0.683$$

The result, 0.683, is the 68% of the 68-95-99.7 rule (see [the 68-95-99.7 rule](#)).

8. How the PDF and CDF are linked

Key point: The area under the PDF gives the CDF, and the slope (G-1823) of the CDF gives the PDF; in calculus terms, integration goes one way and differentiation (G-608) the other.

The two curves of Figure 7 carry the same information:

- **PDF to CDF:** the CDF at x is the area under the PDF up to x (section 7). Collecting area is integration.
- **CDF to PDF:** where the CDF climbs steeply, much probability is packed into a short range, so the density is high. The slope of the CDF at x is the PDF at x . Taking a slope is differentiation.

1. **In words:** the steepness of the CDF at a point equals the height of the PDF there.

2. **Formula:**

$$f(x) = \frac{dF(x)}{dx}$$
$$\frac{dF(x)}{dx} \approx \frac{F(x + 0.5) - F(x - 0.5)}{1}$$

The term $dF(x)/dx$ is the **derivative** (G-595), the exact slope of the CDF at x . The last fraction measures the rise of the CDF over a step of 1 around x .

3. **Example:** for the heights at 165,

$$\frac{F(165.5) - F(164.5)}{1} = 0.0399 = f(165)$$

At 150 the same rise is 0.0130, and $f(150) = 0.0130$: the CDF is flatter there, and the PDF lower.

Figure 8 slides a tangent line along the heights CDF. Watch its slope, printed in the top title: it equals the PDF's height in the bottom panel at every point, largest at 165 and small in both tails.

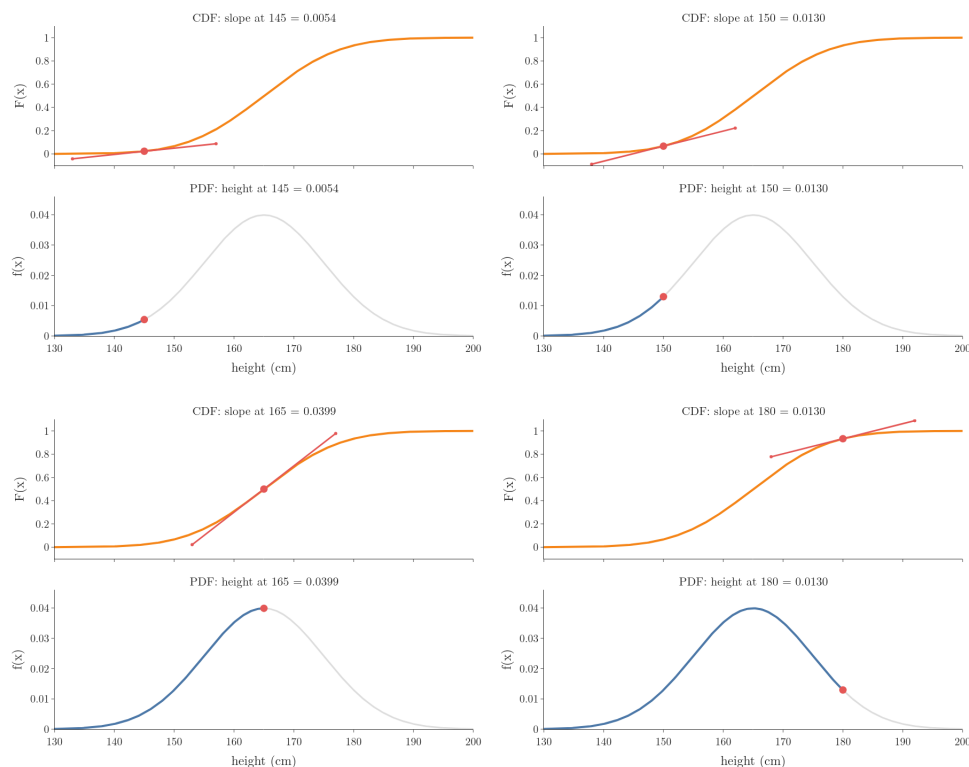


Figure 8: A tangent slides along the CDF of heights; its slope traces out the PDF below

Calculus is not needed to use these ideas: libraries compute both functions. The link returns later: the [z-table](#) is a table of areas under the normal PDF, that is, of its CDF.

9. Summary

Idea	Formula	Example
Probability of one exact value	$P(X = x) = 0$	$P(X = 7.912) = 0$
Probability of a range	$\int_a^b f(x) dx$	$P(8 \leq X \leq 9) = 0.209$
Total area	$\int f(x) dx = 1$	whole CGPA curve
Density as probability per unit	$P(x \leq X \leq x + h) \approx f(x) h$	$0.264 \times 0.01 = 0.00264$
CDF	$F(x) = \int_{-\infty}^x f(t) dt$	$F(150) = 0.067$
PDF from CDF	$f(x) = dF/dx$	slope at 165 = 0.0399

- PMF: discrete, height is probability. PDF: continuous, height is density, area is probability.
- A density can exceed 1; a probability cannot.
- Poisson is discrete (PMF); normal and log-normal are continuous (PDF).
- The CDF of a continuous variable rises smoothly from 0 to 1.

10. Sources

Built from

- CampusX, “Session 40 - Probability Distribution Functions - PDF, PMF & CDF | DSMP 2023”, YouTube, https://www.youtube.com/watch?v=C_QAURbgBqY
- Khan Academy, “Probability density functions”, YouTube, <https://www.youtube.com/watch?v=Fvi9A>
- 3Blue1Brown, “Why ‘probability of 0’ does not mean ‘impossible’ | Probabilities of probabilities, part 2”, YouTube, <https://www.youtube.com/watch?v=ZA4JkHKZM50>

Other references

- SciPy documentation, `scipy.stats.lognorm` and `scipy.stats.poisson`.

11. Key terms

Term	Meaning
Continuous random variable	A random variable that can take any value in a range, with every decimal, such as a CGPA
Probability density function (PDF)	The curve of a continuous random variable whose area over a range is the probability of that range; total area 1
Probability density	Probability per unit of x : the height of a PDF, whose area over a range is a probability
Integration	Finding the area under a curve by adding up infinitely many thin strips

Term	Meaning
$\int_a^b f(x) dx$ (G-13)	The area under f from a to b ; for a PDF, $P(a \leq X \leq b)$
Log-normal distribution	A right-skewed continuous distribution whose logarithm is normal
CDF of a continuous variable	$F(x) = P(X \leq x)$, the area under the PDF to the left of x ; rises smoothly from 0 to 1
Slope	How steeply a curve rises at a point: rise divided by run
Derivative	The exact slope of a curve at each point, written dF/dx
Differentiation	Finding the slope of a curve at each point; the derivative of the CDF is the PDF